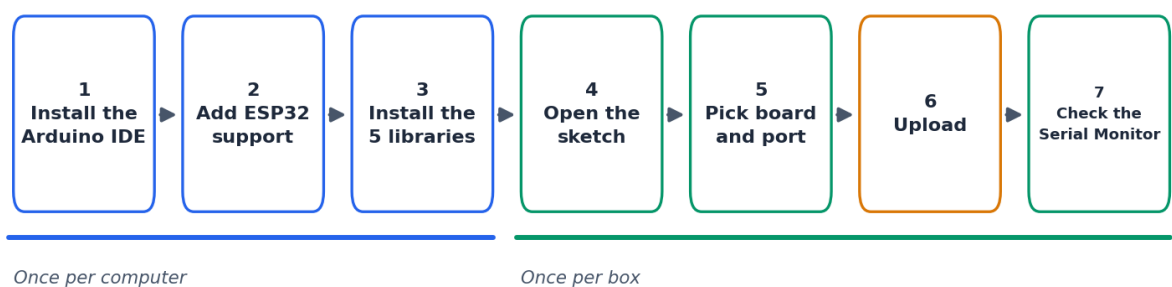


Installing the Firmware

Putting the Magic Boxes firmware onto an ESP32 board with the Arduino IDE

For: Technician / teacher • **Time needed:** About 30 min the first time, 2–3 min per box after that • **Hardware:** ESP32 with VEML7700, BME280, MPU-6500, SCD41

You only need this when building a new box, replacing a board, or updating to a newer firmware version. Boxes that already work do not need reflashing — unless their firmware is too old for an activity you plan to run (see the table below). A Romanian version of this guide is `FLASHING_GUIDE_RO.pdf`.



The first three steps are done once on your computer. The last four are repeated for each box.

About the pictures in this guide

The illustrations are drawings rather than screenshots. The Arduino IDE changes its appearance between versions, so a drawing can point at the thing that matters without going out of date. Menu names are exact; the colours and layout are ours.

Which firmware your activities need

The box prints its version at start-up (FW:, see section 7).

Activities	What the box needs
The Air We Breathe, Indoor Pollution, CO ₂ and Photosynthesis, Air Quality, Greenhouse Effect — anything with CO ₂	An SCD41 CO ₂ sensor, and <code>serial_wifi</code> v2.2.0 or newer (or the USB-only <code>serial_only</code> v1.3.4 or newer). <code>serial_only</code> v1.3.3 and older never read the CO ₂ sensor, even over USB.
Light Is Everywhere, Solar Energy, Solar Panels — outdoors in sunlight	v2.2.5 (or <code>serial_only</code> v1.3.4) or newer. Older firmware stops at about 3800 lux, so every sunny reading looks the same. Above about 10000 lux the absolute value is approximate until checked once against a reference lux meter; comparisons between readings are reliable.
Motion activities near fans or motors	v2.2.5 / v1.3.4 or newer filters out vibration that otherwise shows up as a fake slow wobble.
Everything else	Any version from v2.2.1 (or <code>serial_only</code> v1.3.0) on. Older ones report the motion axes upside down.

The firmware supplied with this guide, `serial_wifi_v2_2_5`, covers all of them, and also works over a USB cable alone: a classroom without WiFi can use it unchanged.

1. What you need

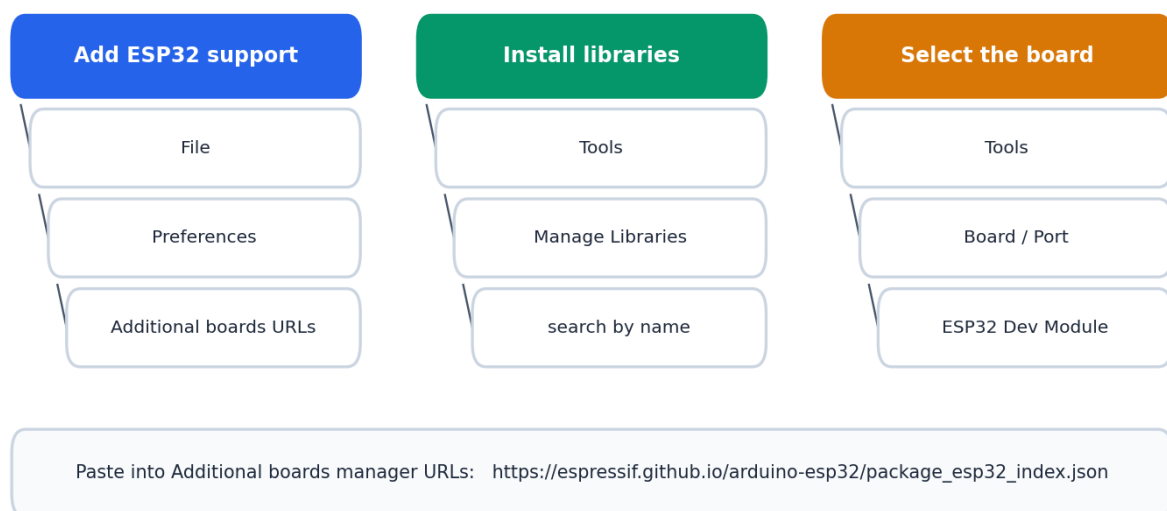
Item	Notes
ESP32 development board	The brain of the box.
USB cable	It must be a data cable. Many phone-charging cables carry power only, and a power-only cable is the most common reason a board never appears in the IDE.
A computer	Windows, macOS or Linux.
The sketch folders	Supplied next to this guide.

2. Install the Arduino IDE

Download it from <https://www.arduino.cc/en/software> and install it. Either version 1.8.x or 2.x works; the menus named below are the same in both.

3. Add ESP32 support and the libraries

The IDE does not know about ESP32 boards until you tell it where to find them, and it does not know the sensors until the libraries are installed.



The three menu paths you need. The long address goes into the Preferences dialogue.

After pasting the address, open **Tools > Board > Boards Manager**, search for **esp32** and install **esp32 by Espressif Systems**. It is a large download, so allow a few minutes.

Then open **Tools > Manage Libraries** and install these five by name, accepting when the IDE offers to add a library's dependencies:

Library	Used for
Adafruit VEMML7700	light sensor
Adafruit BME280 Library	temperature, humidity, pressure
Adafruit Unified Sensor	required by both Adafruit libraries above
ArduinoJson	building the data lines sent to the application
SparkFun SCD4x Arduino Library	CO ₂ sensor

The WiFi, WebServer, DNSServer, Preferences and Wire libraries arrive with the ESP32 board package. Do not install them separately.

4. Open the sketch

The firmware is `serial_wifi_v2_2_5`. It drives all four sensors and speaks to the application over both USB and WiFi.

Open it with **File > Open** and select `serial_wifi_v2_2_5.ino` inside the folder of the same name. Arduino requires a sketch and its folder to share a name, so keep the folder exactly as supplied.

5. Select the board and the port

1. **Tools > Board > esp32 >** your board. If unsure, choose **ESP32 Dev Module**, which suits most common boards.
2. **Tools > Port >** the port that appears when the board is plugged in: `COM5` on Windows, `/dev/cu.usbserial-...` on macOS, `/dev/ttyUSB0` on Linux.
3. Leave everything else at its default. If uploads fail repeatedly, drop the upload speed to 115200.

No port in the menu

The computer is missing the USB-to-serial driver for the chip on your board. Install the **CP210x** driver from Silicon Labs, or the **CH340** driver, depending on which chip the board carries — its documentation will say. Unplug, replug, and look again. If the port still does not appear, try another cable before anything else.

6. Upload

Press **Upload** (the right-pointing arrow). The first compilation of an ESP32 sketch takes a couple of minutes; later ones are much quicker.

Some boards need help entering upload mode:



The button sequence that puts a stubborn board into upload mode.

7. Check that it worked

Open **Tools > Serial Monitor** and set the speed to **115200 baud**. Within a few seconds the box should introduce itself.

```

Serial Monitor — 115200 baud

DEVICE: BOX_3859
TYPE: MULTI
FW: 2.2.5
SENSORS: LUX,WHITE,RAW,T,H,P,ALT,
          AX,AY,AZ,GX,GY,GZ,C02,...
{"id":"BOX_3859","ts_ms":154944,
 "data":{"lux":412.6,"T":26.21,...}}

```

the name of this box
(unique per board)

the firmware version now installed

what was detected — a missing
sensor here is a wiring fault

one line of data per second:
the box is working

A healthy box, and what each line tells you.

Close the Serial Monitor before starting the Magic Boxes application: only one program can hold the port at a time.

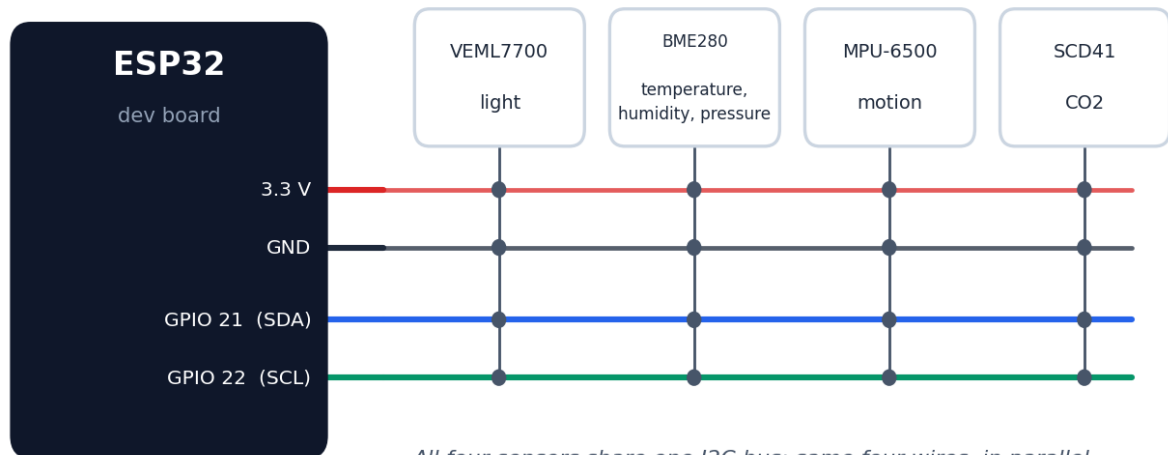
8. Connect the box to WiFi

The network is configured on the box itself, so the sketch never contains a password.

1. On first start, or whenever the saved network is unreachable, the box creates an open WiFi network called **CutiaMagica-XXXX**, where XXXX matches the end of the device name.
2. Connect a phone or laptop to it. A setup page opens automatically; if it does not, browse to **192.168.4.1**.
3. Choose the classroom network, enter its password and save. The box remembers it and reconnects by itself from then on.

The application announces itself on the network every three seconds and the box connects back to it, so no addresses are typed anywhere. USB keeps working at the same time and the application ignores the duplicate data, so a cable can be unplugged mid-lesson without losing the session.

9. Wiring



All four sensors share one I2C bus: same four wires, in parallel.

The bus runs at 100 kHz on purpose. At 400 kHz the CO2 sensor corrupts the bus and the motion sensor disappears.

Four wires, shared by all four sensors.

10. When something goes wrong

Symptom	Likely cause and remedy
No port in the Tools menu Connecting... then a timeout	Power-only USB cable, or a missing CP210x/CH340 driver. Use the button sequence above; try a different USB port or a shorter cable.
Compiles, but the Serial Monitor stays empty	Wrong speed — it must be 115200 baud.
A sensor missing from SENSORS :	A wiring fault, not a firmware one. Check that device's four connections, and that it sits on 3.3 V rather than 5 V.
Values appear, then stop after a while	Update to the current firmware; older versions could be frozen by a stalled sensor.
Light stays at about 3775 lux in bright light	The box runs firmware older than v2.2.5. Reflash it.
No CO ₂ values although the sensor is fitted	The box runs <code>serial_only</code> v1.3.3 or older. Flash <code>serial_wifi_v2_2_5</code> .
The application does not see the box	Close the Serial Monitor, and check that box and laptop are on the same network.
Over WiFi the box never connects, though it is on the right network	The laptop's firewall is blocking the application. Allow it on private networks (TCP port 8765, UDP port 8766). The box retries on its own, at most a minute apart; restarting the application makes it try at once.

Handling

Work on a non-conductive surface and unplug the board before changing any wiring. The sensors run at 3.3 V — connecting one to the 5 V pin can destroy it. Keep boards away from liquids, and hold them by the edges: the pins bend easily and a bent pin is the usual cause of a sensor that has disappeared.

11. Which version am I running?

The box prints **FW:** at boot, and **CHANGELOG.txt** in this folder lists what changed in each version and when. The current firmware is **v2.2.5**.